

Student Dropout Prediction: A Machine Learning Analysis

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Introduction

This study is based on the “Student Dropout and Academic Success” dataset from the UCI repository. Through a complete machine learning pipeline, we aim to predict students' academic outcomes. This presentation will focus on the following two parts:

1. **Clustering Analysis:** We will present the clustering analysis, detailing the process from *algorithm selection* (addressing the limitations of hard boundaries → choosing GMM), *parameter tuning*, to the *validation of clustering result reasonableness*.
2. **Open-ended Exploration:** We will elaborate on the open exploration conducted to enhance predictive performance.

Clustering (1/2): Algorithm Selection & Tuning

1. Hard-Boundary Models

- ▶ **Exploration:** Tested Hierarchical (Ward) and K-Means.
- ▶ **Optimal K:** $K = 4$ provided the best granularity.
- ▶ **Limitation:** While K-Means performed well, rigid spherical boundaries are less flexible in handling the severe structural overlap of the "Enrolled" class.

⇒ *Conclusion: A probabilistic approach is needed to handle the overlap.*

2. Final Selection: GMM

- ▶ **Best Config:** $K = 4$ with **Tied Covariance**.
- ▶ **Why GMM?** Soft-assignments handled overlap naturally; "Tied" covariance reduced 40D overfitting.

Performance Comparison ($K = 4$)

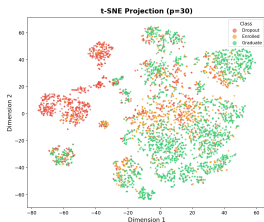
Algorithm	ARI	NMI
Hierarchical (Ward)	0.153	0.149
K-Means	0.313	0.230
GMM (Tied)	0.334	0.273

Clustering (2/2): Core Insight & Visualizations

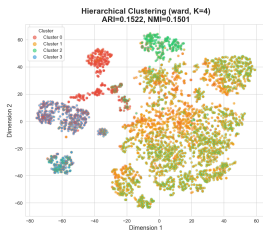
Insight: Natural Label Consistency

- **Consistency:** Clustering results seem to be basically consistent with true labels.
- **Ablation Check:** Removing target-encoded features yielded identical structures.

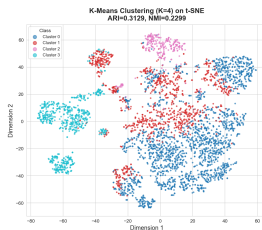
Original Labels



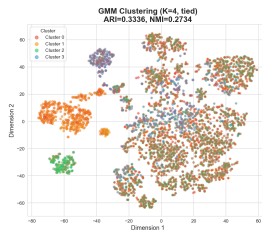
Hierarchical (Ward)



K-Means ($K = 4$)



GMM (Tied)



Open-ended Exploration: LR Class-Bias Tuning

1. Motivation

- ▶ `class_weight='balanced'` overcompensates for Enrolled ($1.86 \times$ weight), causing over-prediction in borderline cases.

2. Method: Boundary Refinement

- ▶ Post-hoc bias: $\hat{y} = \arg \max_k [\log P(y=k | x) + b_k]$
- ▶ Fix $b_{\text{Dropout}}=0$, search only 2 offsets (2-DOF).
- ▶ 41×41 grid on *validation set*.
- ▶ Selected: $b = [0, 0, +0.4]$ — Graduate gets +0.4 log-odds.

3. Significance

- ▶ Bootstrap ($n=10k$): $p = 0.051$ — borderline at 95% level.

4. Model Ranking (Test F1 macro)

Model	Acc	F1
Decision Tree	0.712	0.676
SVM (RBF)	0.732	0.696
Logistic Regression	0.741	0.708
Random Forest	0.747	0.706
ExtraTrees	0.767	0.714
LR + Class-Bias	0.765	0.723

5. Per-Class Gain vs LR

	Drop.	Enr.	Grad.
Δ F1	+0.004	+0.010	+0.025

LR + class-bias tuning: #1 among all 9 models.

Conclusion

Key Findings

1. Engineered top-40 features carry strong signal; academic metrics dominate.
2. **Enrolled** is structurally intermediate — not a natural cluster.
3. LR is the strongest baseline; LR + class-bias tuning achieves best F1 = **0.723**.

Limitations & Future Work

- ▶ LR+bias gain is borderline significant ($p = 0.051$)
- ▶ Future: better Enrolled modeling, external validation, early-warning deployment

Reference

Realhino, V., et al. (2022). Student Dropout and Academic Success. UCI Machine Learning Repository. <https://doi.org/10.24432/C5MC89>

Q & A

Thank You!